Sample for Quiz 2

You may use your notes and the course website.
You may not use Google, any other website, or an LLM.

Load the `nycflights13` package and `flights` data!

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.4.4     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.0
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(nycflights13)
data(flights)
```

1. In `flights` data frame, for every origin, find the number of `NA`'s in each column that is numeric and starts with "arr" or it is character and contains "ca" or "il".

```
flights %>%
  group_by(origin) %>%
  summarise(across(
    .cols = c(where(is.numeric) & starts_with("arr"),
              where(is.character) & contains(c("ca", "il"))),
    .fns = ~ sum(is.na(.x))), .groups = "drop")
```

```
## # A tibble: 3 x 5
##   origin arr_time arr_delay carrier tailnum
##   <chr>     <int>     <int>   <int>   <int>
## 1 EWR        3390      3708       0     606
## 2 JFK        1995      2200       0     909
## 3 LGA        3328      3522       0     997
```

2. In `flights` data frame, for every destination, find the number of flights which have more than 2 hours of arrival delay in Fall season, which are operated by the least popular airline.

```
flights %>%
  group_by(dest) %>%
  filter(arr_delay > 120 & month %in% 7:9) %>%
  group_by(carrier) %>%
  summarise(n = n()) %>%
  arrange(-n) %>%
  slice(1)
```

```
## # A tibble: 1 x 2
##   carrier     n
##   <chr>   <int>
## 1 B6        606
```

3. Determine between which two airports, the worst delayed flights occurred? Note that delayed flights are defined here as `arr_delay > 0 & dep_delay > 0`. How many delayed flights occured between these two airports?

```
flights %>%
  filter(arr_delay > 0 & dep_delay > 0) %>%
  group_by(origin,dest) %>%
  summarise(total_delayed_flights = n(), .groups = "drop") %>%
  arrange(desc(total_delayed_flights)) %>%
  slice(1)
```

```
## # A tibble: 1 x 3
##   origin dest  total_delayed_flights
##   <chr>  <chr>                 <int>
## 1 LGA    ATL                    2839
```

4. Which three carriers have the highest rates of departure delays exceeding 15 minutes? What are the proportions compared to the total number of flights managed?

```
flights %>%
  group_by(carrier) %>%
  summarise(
    n_total = n(),
    n_delay = sum(dep_delay > 15, na.rm = TRUE),
    prop_delayed = n_delay / n_total
  ) %>%
  arrange(desc(prop_delayed)) %>%
  slice_head(n = 3)
```

```
## # A tibble: 3 x 4
```

```
##   carrier n_total n_delay prop_delayed
##   <chr>     <int>   <int>        <dbl>
## 1 EV        54173   15644        0.289
## 2 F9          685     192        0.280
## 3 WN        12275    3250        0.265
```

5. Use `tidyr` parse functions to correct the type of columns in the following data frame. (use their names as a hint)

```
df <- data.frame(
  date = c("2023--11-21", "2023--09-02", "2020--01-30"),
  time = c("01::00-00", "13::58-01", "23::00-09"),
  num = c("10.5", "20.3", "15.8"),
  log = c("TRUE", "N--a", "FALSE"),
  char = as.factor(c("Apple", "Banana", "N-a")),
  fac = c("A", "B", "C"))
df
```

```
##          date      time  num   log   char fac
## 1 2023--11-21 01::00-00 10.5  TRUE  Apple   A
## 2 2023--09-02 13::58-01 20.3  N--a Banana   B
## 3 2020--01-30 23::00-09 15.8 FALSE    N-a   C
```

```
df_edited <- df %>%
  mutate(
    date = parse_datetime(date, format = "%Y--%m-%d"),
    time = parse_time(time, format = "%H::%M-%S"),
    num = parse_number(num),
    log = parse_logical(log, na = "N--a"),
    char = parse_character(as.character(char), na = "N-a"),  # must be converted to character first
    fac = parse_factor(fac)
  )
df_edited
```

```
##         date     time  num   log   char fac
## 1 2023-11-21 01:00:00 10.5  TRUE  Apple   A
## 2 2023-09-02 13:58:01 20.3    NA Banana   B
## 3 2020-01-30 23:00:09 15.8 FALSE   <NA>   C
```